

CYBER SECURITY PROJECT 4

System Vulnerability Checklist & Security Assessment Using Kali Linux

Submitted By: Raja Asim Afraz
Course: Cyber Security
Submission Date: June 2026

Introduction

Cybersecurity plays an important role in protecting computer systems, networks, and sensitive information from cyber threats and unauthorized access. This project focuses on performing a practical system vulnerability assessment using Kali Linux tools and Linux terminal commands. Several cybersecurity techniques were used to inspect firewall status, open ports, running services, localhost vulnerabilities, and software update status.

Tool / Command	Purpose
hostnamectl	Display system information
sudo ufw status	Check firewall status
nmap localhost	Scan localhost for open ports
sudo ss -tulnp	Display open ports and services
apt list --upgradable head -20	Check outdated packages

1. System Information Analysis

Command Used: hostnamectl

```
(raja@raja)-[~]
$ hostnamectl
  Static hostname: raja
            Icon name: computer-laptop
          Chassis: laptop
  System Asset Tag: 5CG423D5W3
        Machine ID: 4d268fdd95f54880bfc8aaeff50f7948
        Boot ID: 4c2116babd8f45ecbfd9bf3938c96cd4
  Operating System: Kali GNU/Linux Rolling
            Kernel: Linux 6.19.11+kali-amd64
        Architecture: x86-64
    Hardware Vendor: Hewlett-Packard
    Hardware Model: HP ProBook 645 G1
    Hardware SKU: D2Z92AV
  Hardware Version: A3009DD10303
  Firmware Version: L72 Ver. 01.35
    Firmware Date: Tue 2015-10-20
    Firmware Age: 10y 7month 2w

(raja@raja)-[~]
$
```

The hostnamectl command was used to gather system information including operating system details, kernel version, hardware architecture, and hostname configuration.

2. Firewall Configuration Analysis

Command Used: sudo ufw status

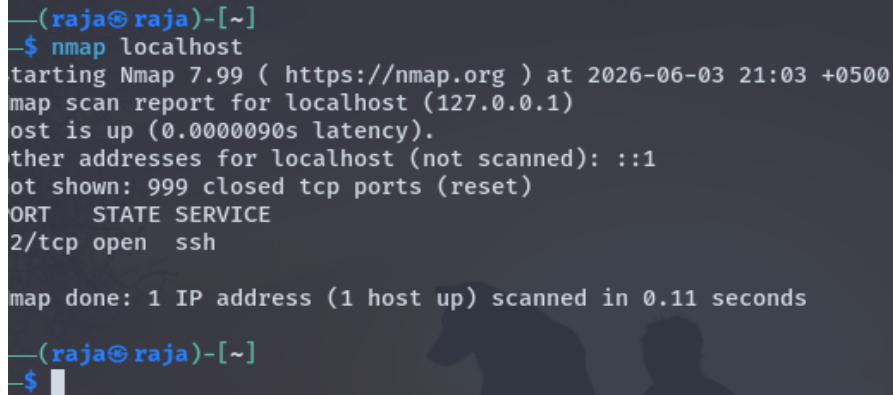
```
raja@raja: ~
(raja@raja)-[~]
$ sudo ufw status
Status: inactive

(raja@raja)-[~]
$
```

The firewall status was checked using the UFW utility. The firewall was found inactive, which may expose the system to unauthorized network traffic.

3. Localhost Vulnerability Scan

Command Used: `nmap localhost`



```
(raja@raja)-[~]
$ nmap localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-03 21:03 +0500
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000090s latency).
No other addresses for localhost (not scanned): ::1
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds

(raja@raja)-[~]
$
```

An Nmap scan was performed on localhost. Port 22 (SSH) was detected in an open state, indicating SSH services were active.

4. Open Ports & Running Services Analysis

Command Used: `sudo ss -tulnp`

```

These make your project look advanced:
—(raja@raja)-[~]
—$ sudo ss -tulnp
etid State Recv-Q Send-Q           Local Address:Port   Peer Address:
ort
dp UNCONN 0      0      Process
dp UNCONN 0      0      224.0.0.251:5353      0.0.0.0:
dp UNCONN 0      0      users:(("brave",pid=88427,fd=21))
dp UNCONN 0      0      *:1716                *:
dp UNCONN 0      0      users:(("kdeconnectd",pid=2040,fd=26))
dp UNCONN 0      0      [fe80::3c38:84fc:bf3e:f0f5]%wlan0:546      [::]:
dp UNCONN 0      0      users:(("NetworkManager",pid=583,fd=31))
cp LISTEN 0      128     0.0.0.0:22            0.0.0.0:
cp LISTEN 0      511     127.0.0.1:35717       0.0.0.0:
cp LISTEN 0      128     [::]:22              [::]:
cp LISTEN 0      50      *:1716                *:
cp LISTEN 0      50      users:(("ssh",pid=753,fd=6))
cp LISTEN 0      50      users:(("code",pid=106110,fd=36))
cp LISTEN 0      128     [::]:22              [::]:
cp LISTEN 0      50      users:(("ssh",pid=753,fd=7))
cp LISTEN 0      50      *:1716                *:
cp LISTEN 0      50      users:(("kdeconnectd",pid=2040,fd=27))

—(raja@raja)-[~]
—$
hostnamectl
↓
—(raja@raja)-[~]
—$

```

The ss command displayed open ports and active listening services including SSH and local communication services.

5. Software Update & Patch Management

Command Used: `apt list --upgradable | head -20`

```

—(raja@raja)-[~]
—$ apt list --upgradable | head -20

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

Listing... and use:
zip/kali-rolling 26.01+dfsg-2 amd64 [upgradable from: 26.00+dfsg1-2]
dduser/kali-rolling 3.155 all [upgradable from: 3.154]
dw-gtk3-kali/kali-rolling 2026.2.0 all [upgradable from: 2025.2.0]
ircrack-ng/kali-rolling 1:1.7+git20230807.4bf83f1a-3 amd64 [upgradable from: 1:
.7+git20230807.4bf83f1a-2+b1]
mass/kali-rolling 5.1.1-0kali1 amd64 [upgradable from: 5.0.1-0kali4]
nt-optional/kali-rolling 1.10.17-1 all [upgradable from: 1.10.15-1]
nt/kali-rolling 1.10.17-1 all [upgradable from: 1.10.15-1]
pache2-bin/kali-rolling 2.4.67-1 amd64 [upgradable from: 2.4.66-8]
pache2-data/kali-rolling 2.4.67-1 all [upgradable from: 2.4.66-8]
pache2-utils/kali-rolling 2.4.67-1 amd64 [upgradable from: 2.4.66-8]
pache2/kali-rolling 2.4.67-1 amd64 [upgradable from: 2.4.66-8]
pparmor/kali-rolling 4.1.7-2 amd64 [upgradable from: 4.1.6-3]
rk/kali-rolling 4:26.04.0-1 amd64 [upgradable from: 4:25.12.3-1]
t-spi2-common/kali-rolling 2.60.3-1 all [upgradable from: 2.60.0-1]
t-spi2-core/kali-rolling 2.60.3-1 amd64 [upgradable from: 2.60.0-1+b1]
tril-common/kali-rolling 1.28.4-1 all [upgradable from: 1.28.2-1]
tril/kali-rolling 1.28.4-1 amd64 [upgradable from: 1.28.2-1+b1]
aloo6/kali-rolling 6.23.0-2 amd64 [upgradable from: 6.23.0-1]
ase-files/kali-rolling 1:2026.1.2 amd64 [upgradable from: 1:2026.1.1]

```

Several upgradeable packages were detected, indicating that multiple software components required security updates.

Risks Identified

1. Firewall protection was inactive.
2. SSH service was exposed through port 22.
3. Multiple outdated packages required security updates.
4. Open listening ports may increase attack surface.
5. Weak firewall configuration may expose network services.

Recommendations

- Enable firewall protection using UFW.
- Regularly update and patch software packages.
- Disable unnecessary network services.
- Monitor open ports regularly.
- Use strong authentication mechanisms for SSH access.
- Perform regular vulnerability assessments.

Conclusion

This project provided practical experience in conducting a vulnerability assessment using Kali Linux tools and Linux terminal commands. Several security aspects including firewall status, open ports, running services, and software updates were analyzed successfully. The project improved understanding of practical cybersecurity assessment techniques and vulnerability management practices.